

# The Ordinal Summit

*Where the Tower Snaps to  $c_1$  | Version 1.3*

ZP Companion | May 2026

This companion explains the ideas in plain language with diagrams and examples. It is not the formal document — every claim here restates a result already proved in ZP-L Incomputability Convergence. Consult that document for the authoritative mathematics.

## What Is ZP-L Doing?

ZP-K proved that the initial machine state  $c_0$  is a Kleene fixed point: a program whose behavior is determined entirely by its own index, guaranteed to exist by the second recursion theorem. ZP-L picks up the thread from a different direction: the ordinal direction. If a map  $\phi$  assigns machine phases to ordinals — sending  $c_0$  to everything below a certain point and  $c_1$  above it — what is that threshold ordinal? ZP-L establishes the answer:  $\epsilon_0$ , the first fixed point of the map  $\alpha \mapsto \omega^\alpha$ .

This is not an arbitrary choice. ZP-L proves that  $\epsilon_0$  is the minimal threshold — no monotone, tower-aligned map can snap before  $\epsilon_0$ , and the canonical map snaps exactly there. ZP-L also establishes the connection to ZP-B: as the tower stages approach  $\epsilon_0$  in ordinal order, their 2-adic encodings converge to  $0 = \perp$  in  $\mathbb{Z}_2$ . The same sequence witnesses both convergences simultaneously.

### Real-world analogy — the unreachable horizon

Imagine a path where each step is longer than the last: you walk 1 mile, then 2, then 4, then 8, ... You are always getting further from the start, but you are also always somewhere on the path — never at the horizon. The horizon ( $\epsilon_0$ ) is the limit of all those steps combined. It exists as a mathematical object, but no finite number of steps reaches it.

The ordinal tower works the same way. The snap that ZP-E proved occurs at  $\epsilon_0$  is not a step you can take from inside the tower. It is the point where the tower stops and something new begins.

## What Is $\epsilon_0$ ?

$\epsilon_0$  (epsilon-zero) is the smallest ordinal  $\alpha$  satisfying the equation  $\omega^\alpha = \alpha$ . In other words, it is the first fixed point of the map that sends any ordinal  $\alpha$  to  $\omega^\alpha$  (omega raised to the power  $\alpha$ ). This is not a coincidence or a definition by fiat — it is a property  $\epsilon_0$  satisfies and no smaller ordinal does.

The tower  $0, 1, \omega, \omega^\omega, \omega^{\omega^\omega}, \dots$  climbs through the ordinals, each stage being  $\omega$  raised to the previous stage. Every stage is strictly below  $\epsilon_0$ , and  $\epsilon_0$  is their limit — the supremum of the whole sequence. It is never reached from below; it is the boundary above all stages.



*Each tower stage is strictly below  $\varepsilon_0$  and maps to  $c_0$ . The snap to  $c_1$  occurs exactly at  $\varepsilon_0$ .*

The tower stages  $0, 1, \omega, \omega^\omega, \dots$  all lie strictly below  $\varepsilon_0$  and map to  $c_0$ . The snap transition to  $c_1$  occurs at the threshold  $\varepsilon_0$  itself. The arrow marks the snap. ZP-L §VII: `epsilon_zero_snap_canonical`.

Remember:  $\varepsilon_0$  is not "infinity." It is a specific countable ordinal — the first one satisfying  $\omega^\alpha = \alpha$ . Ordinals larger than  $\varepsilon_0$  exist ( $\varepsilon_0 + 1, \varepsilon_1$ , and many more). ZP-L is not claiming that  $\varepsilon_0$  is the largest ordinal or that the framework breaks down above it. It is the minimal snap threshold for monotone, tower-aligned maps — and that minimality is exactly what is proved.

## Why $\varepsilon_0$ Is the Exact Threshold

Two facts together pin  $\varepsilon_0$  as the minimal snap threshold. They work like a lower bound and an upper bound that meet at the same point.

Lower bound (snap cannot happen before  $\varepsilon_0$ ): The fundamental sequence is cofinal in  $\varepsilon_0$ . This means: for any ordinal  $\alpha$  below  $\varepsilon_0$ , some tower stage eventually exceeds  $\alpha$ . A monotone map that assigns  $c_0$  to every tower stage must therefore assign  $c_0$  to  $\alpha$  too (it cannot increase between a stage above  $\alpha$  and  $\alpha$  itself). So no ordinal strictly below  $\varepsilon_0$  can be a snap point for such a map.

Upper bound (snap does happen at  $\varepsilon_0$ ):  $\varepsilon_0$  is itself a fixed point of  $\omega^\cdot$  ( $\omega^{\varepsilon_0} = \varepsilon_0$ ). A map that assigns  $c_1$  to all fixed points of  $\omega^\cdot$  must assign  $c_1$  to  $\varepsilon_0$ . Together with the lower bound,  $\varepsilon_0$  is the first place where  $c_1$  can and must appear.

The canonical witness map  $\phi(\alpha) = c_0$  if  $\alpha < \varepsilon_0$ , else  $c_1$  satisfies all five conditions at once — monotonicity, tower-alignment, fixed-point-respecting, snapping at  $\varepsilon_0$ , and minimality — with no free hypotheses. This is the central result of ZP-L.

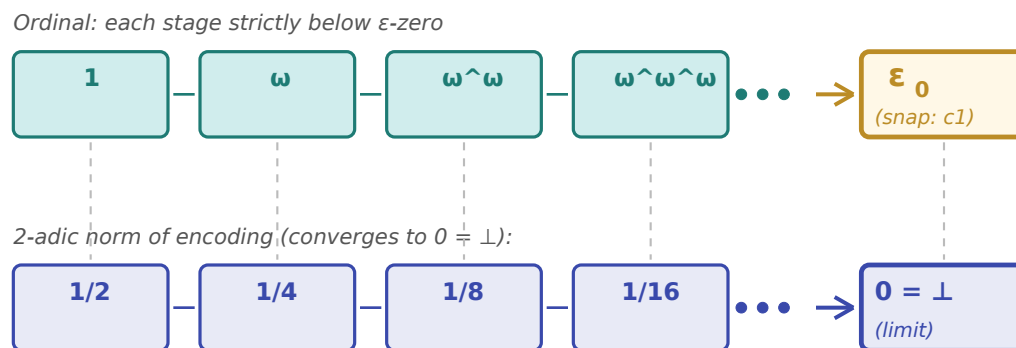
"Minimal" does not mean "unique." ZP-L proves that no snap can occur strictly before  $\varepsilon_0$  (under the stated conditions), not that  $\varepsilon_0$  is the only possible snap point. Maps that also assign  $c_1$  to ordinals above  $\varepsilon_0$  are not ruled out. The framework fixes  $\varepsilon_0$  as the threshold by the canonical witness, not by uniqueness of all satisfying maps.

## The Two-Adic Connection

ZP-B established that  $0 = \perp$  in the 2-adic integers  $\mathbb{Z}_2$  is the valiative sink: the 2-adic valuation of 0 is infinite, which means 0 sits at the bottom of the 2-adic metric. ZP-L connects this to the ordinal tower via an

encoding of ordinals below  $\varepsilon_0$  into  $\mathbb{Z}_2$  using their Cantor normal forms.

The encoding (`cnfToZp2` in `ZPL.lean`) maps the  $n$ -th tower stage to  $2^n$  in  $\mathbb{Z}_2$ . The 2-adic norm of  $2^n$  is  $(1/2)^n$ , which goes to 0 as  $n$  increases. So as the ordinal tower climbs toward  $\varepsilon_0$  from below, its 2-adic image converges to  $0 = \perp$  in the 2-adic metric (the norms  $(1/2)^n$  decrease monotonically and converge to 0). Two convergences, one tower.



Same tower, two perspectives. Teal (top): ordinal stages approach  $\varepsilon$ -zero. Indigo (bottom): their 2-adic norms approach 0.

Same tower stages, two perspectives. Teal (top row): ordinal stages  $1, \omega, \omega^\omega, \dots$  approach  $\varepsilon_0$  from below. Indigo (bottom row): the 2-adic norms of their encodings  $1/2, 1/4, 1/8, \dots$  converge to  $0 = \perp$ . ZP-L §VII: `snap_zp2_correspondence`.

This dual convergence is what `snap_zp2_correspondence` formalizes in ZP-L §VII: four jointly established facts about the same tower sequence — all stages below  $\varepsilon_0$  in ordinal order, all assigned  $c_0$  by the canonical map, all encoding to  $2^n$  in  $\mathbb{Z}_2$  with 2-adic norm  $(1/2)^n \rightarrow 0$ , and the canonical map snapping to  $c_1$  at  $\varepsilon_0$ .

## The Kleene Connection

ZP-K identified  $\perp$  as a Kleene fixed point: a program that is its own output, guaranteed to exist by Kleene's second recursion theorem. ZP-L identifies  $\varepsilon_0$  as an ordinal fixed point: the first ordinal satisfying  $\omega^\alpha = \alpha$ , guaranteed to exist by the fixed-point theorem for normal functions. Both fixed-point proofs require `Classical.choice` in their Lean formalizations. In each case, the choice axiom enters at a non-constructive selection step: the ordinal proof uses it for the supremum that defines  $\varepsilon_0$ ; the Kleene proof uses it for the fixed-point selection. Whether these reflect the same underlying mathematical phenomenon — or are coincidentally parallel — is the open question ZPL.lean §I poses.

ZP-L §VI (`kleene_ordinal_snap_bridge`) names this structural parallel explicitly. The theorem itself is purely ordinal — no `Code` or `eval` object appears in it. What the section establishes is that the hypothesis structure of the two proofs is the same: diagonalization, non-constructive selection, fixed-point existence.

**Two rooms, same pattern**

Imagine two mathematicians working in separate rooms. One (ZP-K) is computing with programs and asking: what program runs itself? The other (ZP-L) is computing with ordinals and asking: what ordinal satisfies  $\omega^\alpha = \alpha$ ? Neither knows what the other is doing. When they compare notes, they find the same proof structure: the same diagonal argument, the same non-constructive selection step, the same kind of fixed-point existence result.

ZPM formalizes this parallel with a type bridge between the two settings. The two rooms share the same floor plan.

## Convergence with Proof Theory

ZP-L derives  $\varepsilon_0$  as the snap threshold from ordinal fixed-point structure alone —  $\omega$ -tower iteration, the fixed-point property of  $\alpha \mapsto \omega^\alpha$ , and Cantor normal form encodings into  $\mathbb{Z}_2$ . Remark R-L.1 in the formal document notes a structural alignment with an independent result from proof theory.

Gentzen (1936) proved that  $\varepsilon_0$  is the proof-theoretic ordinal of Peano Arithmetic: PA can prove transfinite induction up to any ordinal strictly below  $\varepsilon_0$ , but not for  $\varepsilon_0$  itself.  $\varepsilon_0$  is the exact ordinal boundary of PA's own proof machinery — the furthest point its provability reaches before requiring a strictly stronger system.

ZP-L and Gentzen arrive at  $\varepsilon_0$  from entirely separate starting points. ZP-L starts from ordinal fixed-point structure:  $\varepsilon_0$  is where  $\omega$ -tower self-iteration closes on itself. Gentzen starts from proof theory:  $\varepsilon_0$  is where PA's provability tower runs out. Neither derivation references the other's domain. Both locate the same boundary.

The convergence is notable. ZP-L's fixed-point derivation and Gentzen's proof-theoretic analysis were conducted in separate domains with separate machinery. That both locate  $\varepsilon_0$  is a structural observation, not an argument for either result. Both stand or fall on their own proofs.

ZP does not prove any part of Gödel's incompleteness theorems, and does not reprove Gentzen's ordinal analysis of PA. The connection is a structural observation: two independent formal derivations arrive at the same ordinal boundary. Gentzen (1936) established that  $\varepsilon_0$  is the specific ordinal where PA's proof-theoretic strength is exhausted. ZP-L established  $\varepsilon_0$  as the snap threshold from ordinal fixed-point structure. No formal equivalence between the two derivations is claimed.

## A Note on Classical.choice

Every ZP-L theorem carries the axiom footprint [propext, Classical.choice, Quot.sound]. This is the standard footprint of Lean 4 + Mathlib for any theorem that uses ordinal theory, p-adic analysis, or computability. It is not a novel ZP-L commitment.

What is notable is where Classical.choice appears. Across the four mathematical settings of the ZP framework — topology (ZP-B), information theory (ZP-C), set theory and computation (ZP-J/K), and ordinal theory (ZP-L) — the same axiom appears at the same structural step: the non-constructive diagonal. ZP-L §I documents this convergence. Whether Classical.choice is structurally forced by the ZP framework — rather than merely inherited from Mathlib infrastructure — remains an open question.

Key Result — ZP-L: The Ordinal Snap Threshold

epsilon\_zero\_snap\_canonical (ZPL.lean §VII): The canonical map  $\phi(\alpha) = c_0$  if  $\alpha < \varepsilon_0$ , else  $c_1$  satisfies all five conditions simultaneously — monotone, tower-aligned ( $c_0$  on every fundamental sequence stage), fixed-point-respecting ( $c_1$  on every fixed point of  $\omega^\cdot$ ), snapping at  $\varepsilon_0$ , and  $\varepsilon_0$  minimal — with no free hypotheses. snap\_zp2\_correspondence co-proves that the same tower witnesses both the ordinal approach to  $\varepsilon_0$  and the 2-adic convergence to  $0 = \perp$ . All theorems proved without sorry. ZPM formalizes the type bridge between the ordinal and machine-phase encodings. ✓